

(335) C++ Programming

Description & Eligibility

Evaluate knowledge of working with structured designs, algorithms, and OOP methodology using the C++ language. This contest can only be completed using the C++ programming language. This contest will be graded on the Windows operating system; therefore, Unix/Linux should *not* be used. Any postsecondary or secondary division contestant may enter this application event.

Contestant Must Supply

- Published and/or unpublished non-electronic written reference materials
- Computer or laptop/notebook (USB-Type A or adapter required); a full-size keyboard may be used (no printer is needed) and supporting equipment (extension cords, etc.)
- Contestant must use one of these IDEs: Visual Studio 2022, Visual Studio Code, DevC++ with Ming compiler, Code::Blocks
- Only pre-written code that is hard copied is allowed

Competition Notes

- Method of evaluation: application.
- Carry-in and setup of equipment must be done solely by the contestant within the time allotted.
- All student-produced jobs completed on-site are uploaded electronically and are not printed at the national level.
- Business Professionals of America assumes no responsibility for hardware/software provided by the contestant.
- No equipment, supplies or materials other than those specified for the event will be allowed in the testing area.
- Previous Business Professionals of America tests and/or sample tests (handwritten, photocopied or keyed) may not be taken into the testing area. **Violation of this rule will result in disqualification.**
- Length of event: no more than ten (10) minutes orientation, no more than ninety (90) minutes testing time, and no more than ten (10) minutes wrap-up.
- Entries vary by state.

Contest Competencies

- Demonstrate understanding of general programming concepts and C++ computer language
- Use programming skills for proper development of a C++ computer program
- Demonstrate knowledge of C++ computer language
- Implement program logic (algorithms, structured design)
- Use structural design techniques and object-oriented concepts
- Understand and implement UML diagrams
- Create a C++ program using calculations, totals, selection statements, logical operators, classes, sequential file access, I/O operations, loops, methods, arrays, and data structures (linked lists, structures, etc.)